



Contact

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Skills

Analytics & Visualisation:

Python, Power BI, MS-Excel

Data Science Libraries:

Pytorch, Keras, sklearn, pandas,
pytorch

Deployment, Containerisation

Flask, Streamlit, Docker

DBMS:

MySQL, PostgreSQL, MongoDB,
MariaDB

Certifications

Google IT Support Certification,
Introduction to Linux,
SQL for Data Science,
Machine Learning Foundations: A
Case Study Approach

Teaching

Teaching Assistant, NPTEL

(Jan 2024- April 2024)
noc24-cs47:
The Joy of Computing Using Python

Student Mentor, IITM BS Degree Programme

(Mar 2024- April 2024)
BSMA1004:
Statistics for Data Science 2.

Volunteering

COVID-19 Response Volunteer:

Caregiver Saathi, May 2021

Others

.GATE DA 2025: Rank 3876
(March 2025)

Winner Hack Kochi #2:

(November 2019)
IEEE ORE, Kochi Hub

KEAM- Engineering: Rank 333
(August 2018)
Kerala State Entrance Examination

Highest Scorer: Science Stream:
(May 2018)

Maria Thurkadayil

DATA SCIENTIST

Summary

Master's Degree Student building skills in the field of Data Science, Prior background in Computer Science and Engineering. Looking for opportunities to work with latest technologies

Education.

Chennai Mathematical Institute,
Data Science, Aug 2024 - May 2026

Master of Science

Experience

Graduate Summer Intern, S3 Labs, IIT Bhilai | May 2025- July 2025

- Technical Image Understanding**
 - Evaluating how Packet Diagrams, Sequence Diagrams, Class Diagrams, State Diagrams, etc are inferred using LLM's. using a set of hand drawn images and mermaid code diagrams
- Legal Infographic Generation** Dataset : IL-TUR(Exploration Lab, IIT Kanpur)
 - Conducted Research Paper Analysis on the Applicability of LLM's in the Legal Domain
 - Generated trial prototypes for generating Infographics like Timelines, Key Arguments**Supervisor:** Gagan Raj Gupta, Department of Computer Science and Engineering

Projects

- Forecasting the Total Marine Fish Production In Kerala -
Development & Comparative Analysis of Machine Learning Models**
Machine Learning, Fisheries | Jan 2022- May 2022
Technology : SPSS, scikit-learn, pandas
 - Marine Fish production Data in Kerala (in tonnes) was collected from **CMFRI, Kochi** and data on 9 significant environmental variables from **ICOADS** from 1960 - 2014 .
 - Highest predictive power was for Random Forest- R2 score 0.87, MAE 0.05, MSE 0.005, RMSE 0.07
- Prediction of Crime Categories**
Ensembled Learning, Law & Order | June 2024–August 2024
Technology: xgboost, lightgbm, pandas, scikit-learn
 - The dataset consisted of 20,000 rows & 20 columns about location, victim demographics, date, time, etc. 6 crime categories were possible for the target variable.
 - Data Exploration, Cleaning and Preprocessing created 23 relevant feature columns.
 - LGBMClassifier()** and **XGBoostClassifier()** were used in building a **VotingClassifier()** for which had an accuracy of 0.88080 in contrast to the baseline accuracy of 0.58.
- Perspective Aware Sentiment Analysis**
NLP, Sentiment Analysis | November 2025
Models: Roberta-base, GPT-4o-mini
Technology: transformers, openai, feedparser, pandas, python-dotenv
 - Real-time data pipeline using **Google News RSS** to fetch and parse headlines based on specific search queries and time windows.
 - Implemented a standard sentiment baseline using the **cardiff nlp/twitter-roberta-base-sentiment-latest** model to provide objective scores. Perspective-aware sentiment engine was created using GPT-4o-mini that adapts its analysis based on user-defined personas .
- Differential Privacy in PyTorch (Opacus)**
Open Source Contribution, Data Privacy and Data Quality Project | May 2026
 - Attempted a Contribution to Opacus, Meta's open-source library enabling differentially private deep learning with minimal code changes.
 - Worked on an issue that had a privacy bug (silent epsilon miscalculations with WeightedRandomSampler).
- Synthetic Data Generation Pipeline — MIRA, IIT Jammu (Ongoing)**
 - Worked on a Python pipeline for generating instruction fine-tuning data (context, instruction, response triples) from raw text documents using self-hosted LLMs
 - Set up and validated the local development environment, connecting to MIRA's private GPU inference cluster (Gemma 4, GPT-OSS 20B) over Tailscale VPN